AP CALCULUS AB

Unit 1 • Lesson 1.16 practice

Lesson 1.16 • Working with the Intermediate Value Theorem

4 multiple-choice questions and 1 free-response question

Question 1

Which hypothesis is required to apply the Intermediate Value Theorem to f on $[a, b]$?

(A) f is continuous on $[a, b]$.
(B) f is increasing on $[a, b]$.
(C) $f(a)$ and $f(b)$ have the same sign.
(D) f is a polynomial.

Question 2

f is continuous on $[0, 4]$ with $f(0) = 3$ and $f(4) = -1$. Which must be true?

(A) $f(2) = 1$
(B) $f(c) = 0$ for some c in $(0, 4)$
(C) f has exactly one zero in $(0, 4)$.
(D) $f(c) = 4$ for some c in $(0, 4)$

Question 3

For $f(x) = \frac{1}{x}$, $f(-1) = -1$ and $f(1) = 1$, yet $f(x) \neq 0$ for every x. Why does this not contradict the IVT?

(A) 0 is not between -1 and 1.
(B) The IVT only applies to polynomials.
(C) f is not continuous on $[-1, 1]$.
(D) f is decreasing.

Question 4

g is continuous with $g(0) = 2$, $g(1) = -1$, $g(2) = 1$, $g(3) = 4$. What is the least number of zeros g must have on $[0, 3]$?

(A) 0
(B) 1
(C) 3
(D) 2

Question 5 (Free response • calculator for (c))

Let $f(x) = x^3 - 2x - 2$.

(a) Use the IVT to show that f has a zero in $[1, 2]$.
(b) Show that $f(c) = 10$ for some c in $[2, 3]$.
(c) Use a calculator to approximate the zero of f in $[1, 2]$ to three decimal places.
(d) Explain why the IVT cannot be applied to $g(x) = \dfrac{1}{x - 1.5}$ on $[1, 2]$.